

M1.3 — Bucket 2 claims

Live working Markdown for Bucket 2 claim wording, argument, and probe/control evidence. Claims here need new fixtures, controls, or interventions beyond the locked M12x Stage-3 package. Refine one at a time; none are locked until their write-up is complete. TeX companion: `docs/report/findings/milestone1_part3_bucket2_claims.tex`.

1 Claim 1 — Causal-role underdetermination under learner-input equivalence

Claim. Let $\mathcal{T} = (\mathcal{F}_{\text{BK}}, E^+, E^-, C)$ denote the learner-visible task for a fixed configuration C . If two causal designs (G, t) and (G', t) induce the same $\mathcal{T}$, but assign the direct-parent role to different extensionally equivalent predictors, unguided ABA Learning has no learner-visible basis for preferring either predictor on causal grounds. The formal learning problem is the same under both graphs, and the implemented search receives the same input. Any preference between the predictors must therefore arise from representation or search bias rather than from evidence about their causal roles. Consequently, the same fully covering framework can be mechanism-aligned under one graph and misaligned under the other.

Argument. An ABA Learning task is specified by an initial ABA framework, positive and negative examples, and a reasoning mode. Its solution condition concerns acceptance of E^+ and rejection of E^- . It does not refer to G , $\text{Pa}(t)$, or the distinction between a parent and an extensionally equivalent non-parent. The current tabular pipeline preserves this separation: it serialises non-target columns into exact-value BK, while handcrafted graph edges are used only for graph-relative evaluation.

M13-C1 demonstrates the consequence using the shared table

$$\mathcal{D} = \{(x_0, x_1, x_2) = (0, 0, 0), (1, 1, 1), (2, 2, 2)\},$$

with target x_2 , $E^+ = \{x_2(2), x_2(3)\}$, and $E^- = \{x_2(1)\}$. Under G_0 , x_0 is the common cause of x_1 and x_2 , so $\text{Pa}_{G_0}(x_2) = \{x_0\}$. Under G_1 , x_1 is the common cause of x_0 and x_2 , so $\text{Pa}_{G_1}(x_2) = \{x_1\}$. The ordered table, BK, target, examples, and configuration are identical. Stage-0 tests and the generated comparison confirmed byte-identical data and BK, identical examples, identical normalised learned δ , and identical ASP coverage within both configuration pairs.

Under ECAI, both graph-labelled cells learned

$$x_2(A) \leftarrow x0_val_1(A), \quad x_2(A) \leftarrow x0_val_2(A).$$

The traces show one-token non-deterministic folding selecting the first matching x_0 predicates; no assumptions or contraries were introduced. These rules are an exact parent-based solution under G_0 , but use only the extensionally equivalent sibling under G_1 . Nevertheless, both cells cover all 2/2 positives and reject the 1/1 negative. Thus the same learned framework changes

from `clean_recovery=1` to `clean_recovery=0` solely because the external graph-relative interpretation changes.

Under AAMAS, both cells learned

$$\begin{aligned}x_2(A) &\leftarrow x0_val_1(A), x1_val_1(A), \\x_2(A) &\leftarrow x0_val_2(A), x1_val_2(A).\end{aligned}$$

Greedy folding accumulated both matching predictors for each positive row. The learned framework therefore contains the parent and the extensionally equivalent sibling under either graph, with variable-level parent precision $1/2$, recall 1, and full ASP coverage. It expresses no preference between their causal roles.

The new paired control isolates what the M12x results could only illustrate. Bucket 1 showed that ECAI’s x_0 -keyed rules and AAMAS’s full-row conjunctions can cite non-parents while retaining sample separation. M13-C1 holds the complete learner-visible task fixed and changes only the valid causal interpretation. The resulting graph-invariant learned frameworks confirm that coverage cannot resolve causal role when the relevant distinction is absent from the learning input. This is an information limit of the unguided task; it does not imply that ABA Learning cannot recover a true parent when the available representation and search happen to select it.

Evidence. Experiment record: `docs/experiments/qualitative/M13-C1-causal-role-underdetermination`.
Generated comparison: `causal/outputs/aba_learning/grid/M13_c1_role_equivalence_summary.md`.

2 Claim 2 — Configuration-dependent sensitivity to BK feature-block order

Claim. Under the published configurations on the `val`-only M13-C2 controls, complete predictor-block permutations changed ECAI’s normalised learned framework in every tested representation family, while AAMAS preserved one normalised framework and one ASP-coverage result per family. For ECAI, the leading feature block determined the variable used in the ordinary target-rule backbone, and subsequent block order determined the assumption and contrary repair. This changed intensional recovery on U2 and U5, and changed both intensional recovery and negative rejection on U7. For AAMAS, BK order changed the greedy traversal sequence but not the sorted row-wise conjunctions. This is a bounded claim about the exact-value U2, U5, and U7 controls; it is not a claim that greedy ABA Learning is order-invariant for arbitrary ABA frameworks.

Argument. ABA Learning theory allows the order of Folding and Assumption Introduction to affect the learned framework. The implementation supplies a specific representation channel: generated BK follows predictor-column order, BK rules receive identifiers in file order, and folding enumerates those identifiers through the folding lookup table. Under ECAI, the initial token budget permits a one-literal fold. The first successful predicate is retained if it passes entailment; otherwise Assumption Introduction preserves that body and contrary learning continues from the remaining ordered folding material. Under AAMAS, greedy folding instead visits every matching original-BK rule for the positive row, accumulates their heads, and sorts the result.

M1.1 had already established the ECAI-side mechanism under a related non-deterministic configuration. Its categorical-three σ -controls and ABL-101 intervention linked BK order to the first fold, entailment result, and assumption commitment. M12x then supplied the fixed-order patterns: all nine ECAI cells used x_0 in their ordinary target bodies, whereas all nine AAMAS cells accumulated every non-target exact-value predicate true on each positive row. Those results motivated an order intervention but did not themselves perform one.

M13-C2 held the graph, named values, row identifiers, target, $E^\pm$, and the set of BK rules fixed within each family. Only complete predictor blocks were permuted. Stage-0 tests verified these invariants. Under ECAI, the two U2 orders produced two distinct deltas, the two U5 orders produced two distinct deltas, and all six U7 orders produced distinct deltas. In every cell, the first target fold used the leading feature block. Under AAMAS, the two U2 orders, the two U5 orders, and all six U7 orders each produced one normalised delta for their family, with unchanged assumptions and coverage.

U5 gives the clearest semantic contrast. With $x_0 \prec_{\text{BK}} x_1$, ECAI reproduced the M12x framework: ordinary target rules were keyed by the non-parent grandparent x_0 , two assumptions were introduced, and x_1 occurred in contrary rules. The framework fully separated the examples but was intensionally misaligned. Reversing the blocks to $x_1 \prec_{\text{BK}} x_0$ produced

$$x_2(A) \leftarrow x1_val_1(A), \quad x_2(A) \leftarrow x1_val_2(A).$$

These folds passed entailment immediately. The learned framework contained no assumptions or contraries, exactly matched $\mathcal{H}_{x_2}^*$, and achieved clean parent recovery. AAMAS returned the same four parent-plus-grandparent conjunctions under both orders.

U2 supplies a both-parent control. Reversing its two parent blocks changed whether ECAI used x_0 or x_1 in the ordinary target bodies and exchanged which parent appeared in the contrary rules. Both frameworks remained semantically successful encodings of both-parents-nonzero with full coverage. AAMAS returned the same exact four both-parent conjunctions under both orders. Thus ECAI’s representation changed even where the semantic verdict did not.

U7 shows that the same order dependence can reach coverage. ECAI’s ordinary target bodies always used the first block. When x_2 was third—under $x_0 \prec x_1 \prec x_2$ or $x_1 \prec x_0 \prec x_2$ —contrary repair remained within $\{x_0, x_1\}$, introduced six assumptions, omitted x_2 , and rejected 0/3 negatives. Every order placing x_2 in the first two blocks introduced only three assumptions and rejected all 3/3 negatives. The parent-only orders $x_1 \prec x_2 \prec x_0$ and $x_2 \prec x_1 \prec x_0$ yielded assumption-mediated frameworks over both direct parents and expressed the intended $x_1 \neq x_2$ mechanism without citing x_0 . Orders pairing x_0 with x_2 achieved full coverage through the ancestor proxy but remained intensionally misaligned.

Across the same six U7 orders, AAMAS always learned the same three x_0, x_1, x_2 conjunctions, introduced no assumptions, and retained full coverage. Its trace visited feature predicates in BK order, but greedy accumulation and final sorting removed that order from the normalised learned rules. The comparative result is therefore not merely that the algorithms print literals differently. In these controls, BK order selected materially different ECAI frameworks and sometimes changed whether they covered the examples, whereas it did not change the normalised AAMAS outcome.

Evidence. Experiment record: docs/experiments/qualitative/M13-C2-bk-feature-order/experiment.md.
Generated comparison: causal/outputs/aba_learning/grid/M13_c2_bk_order_summary.md.